MTH2003 Feedback Sheet 1 Solutions

1. (a) **Question** $t\dfrac{\mathrm{d}y}{\mathrm{d}t} + y + 4t = 0.$

Solution

[i] The standard form of the equation is $\dfrac{\mathrm{d}y}{\mathrm{d}t} = -4 - \dfrac{y}{t} = -\dfrac{4t+y}{t}$ and the variables do not separate. To confirm, we can take $a = b = 1$, $c = d = 2$ and find $f(a,b)f(c,d) = 25$, while $f(a,d)f(c,b) = 27$.

[ii] This equation is homogeneous, as is evident from the form $\dfrac{\mathrm{d}y}{\mathrm{d}t} = -4 - \dfrac{y}{t}$. As usual, put $y = tv$, $\mathrm{d}y/\mathrm{d}t = t\mathrm{d}v/\mathrm{d}t + v$ which leads to $t\mathrm{d}v/\mathrm{d}t = -2(2+v)$. This is a separable equation. We have $\mathrm{d}v/(2+v) = -2\mathrm{d}t/t \Rightarrow \ln(|v+2|) = -2\ln(|t|) + C_1 \Rightarrow |v+2| = C_2|t|^{-2} \Rightarrow v = -2 + C_4 t^{-2}$. Substituting this into $y = tv$, we get ultimately $\boxed{y = -2t + C_4 t^{-1}}$, defined for $t \in (-\infty, 0)$ and $t \in (0, \infty)$. The special solution of the separable equation is $v = -2$, giving $y = -2t$, which is covered by the general formula for $C_4 = 0$

[iii] We can write $\mathrm{d}y/\mathrm{d}t + y/t = -4$, and the equation is linear, with $p(t) = 1/t$, $g(t) = -4$. Hence $\int p(t)\,\mathrm{d}t = \ln|t| + c_1$, and the integrating factor can be chosen as $h(t) = \exp[\ln(t)] = t$. Multiplying the equation by t we have $ty' + y = -4t \Rightarrow (ty)' = -4t$, wherefrom $ty = -2t^2 + C$ and $\boxed{y = -2t + C/t}$, defined for $t \in (-\infty, 0)$ and $t \in (0, \infty)$ as before.

2. (a) **Question** $y' = -4/y$, *(A)* $y(1) = 1$, *(B)* $y(-1) = -1$.

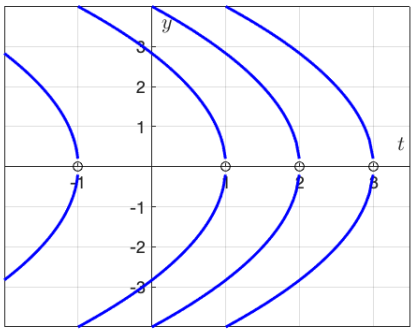
Solution

[i] The right-hand side is $f(t,y) = -4/y$, which is continuous for all (t,y) except when $y = 0$. The partial derivative $f_y(t,y) = 4/y^2$ is continuous for the same arguments. Hence the IVP will be locally solvable for any pair of (t_0, y_0), where $\boxed{y_0 \neq 0}$ and the solution will be locally unique subject to the same condition.

[ii] Possible factorisation: $F(t) = 1$, $G(y) = -4/y$. Equation $G(y) = 0$ does not have roots so no special solutions. Typical solutions: $y\,\mathrm{d}y = -4\,\mathrm{d}t$, $\frac{1}{2}y^2 = 4C - 4t$, $\boxed{y = \pm\sqrt{8(C-t)}}$. Each is defined as long as the expression under the square root is positive, that is, for $\boxed{t \in (-\infty, C)}$, and is unique in that interval.

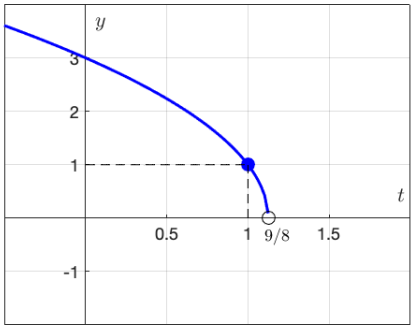
[iii]

The integral curves have the shape of parabolas, "looking leftwards".
The summits of these parabolas are at their crossings of the t-axis, and should be excluded from solutions since y is in the denominator of the right-hand side, so the equation is not defined at these points.



[iv]

(A) We substitute $t = 1$, $y = 1$ into the typical solution and get $1 = \pm\sqrt{8(C-1)}$, hence $C = 9/8$ and we must take the upper sign. The solution is therefore $\boxed{y = \sqrt{9-8t}}$. The argument of the square root must be non-negative and $y \neq 0$, so the solution is defined in the interval $\boxed{I = (-\infty, 9/8)}$ and is unique in that interval.



(B) We substitute $t = -1$, $y = -1$ into the general solution and get $-1 = \pm\sqrt{8(C+1)}$, hence $C = -7/8$ and we must take the lower sign. The solution is therefore $\boxed{y = -\sqrt{-7-8t}}$. The argument of the square root must be non-negative and $y \neq 0$ so the solution is defined in the interval $\boxed{I = (-\infty, -7/8)}$ and is unique in that interval.

